

Risk-Calibrated Fallback: A Selective-Prediction Approach to Minimum-Risk-Manoeuvre Triggering in Closed-Loop Autonomous Driving

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Abstract—Every deployed autonomous vehicle has a minimum-risk manoeuvre (MRM) — pull over, slow down, hand control back to a driver — and when to trigger it is almost universally a hand-tuned threshold on a confidence score. That score is precisely the quantity known to be untrustworthy under distribution shift, so the trigger for the safety behaviour rests on a calibration that can silently expire exactly when the vehicle needs it most. Prior conformal-prediction work in this space is almost entirely open-loop: it produces prediction regions and stops, so it cannot see that a conservative region drives an MRM trigger, and that triggering itself has a safety cost — freezing in a live lane. We formalise MRM triggering as a selective-prediction problem with a guarantee on the *decision*, not merely on the prediction, using Learn-then-Test to certify a trigger threshold against a bounded miss-rate target under exchangeability, and Conformal Risk Control as a tighter alternative when the closed-loop loss is empirically monotone in the threshold. We evaluate in closed loop on real, replayed driving scenarios (Argoverse 2, nuScenes) under reactive traffic, using a fallback manoeuvre compliant with UN Regulation No. 157’s minimum-risk-manoevrue deceleration bound and harm labelled by the nuPlan closed-loop safety definition (at-fault collision or time-to-collision below 0.95 s). We report both the missed-intervention rate the calibrated trigger is built to bound and the unnecessary-stop rate open-loop evaluation cannot see, and we show that the two are inseparable: any trigger design that only reports one is reporting half the picture.

Index Terms—Autonomous driving, conformal prediction, selective prediction, minimum-risk manoeuvre, closed-loop safety evaluation, trajectory prediction, risk calibration, distribution shift.

I. INTRODUCTION

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk manoeuvre: slow to a stop, pull to the shoulder, or otherwise hand control back to a human. The decision of *when* to invoke this behaviour is, in nearly every deployed and published system, a hand-tuned threshold on some scalar confidence or uncertainty score produced by a perception or prediction module. That practice has a specific and largely unexamined weakness: the very quantity the threshold is applied to — a learned model’s uncertainty estimate — is exactly what has been shown to degrade under the kind of distribution shift a vehicle experiences when it enters a new city, a new weather condition, or simply a stretch of

road unlike anything in its calibration set. A threshold tuned under one operating distribution carries no guarantee once that distribution has moved, and the failure is silent: the system continues to report a confidence number, and nothing in the number itself signals that the number is now wrong.

Conformal prediction offers a principled alternative: rather than a point confidence score, it produces a prediction region with a distribution-free, finite-sample coverage guarantee under exchangeability. A substantial and fast-growing literature has applied conformal prediction to trajectory forecasting and to planning under uncertainty in autonomous driving. Almost without exception, however, this literature is *open-loop*: it constructs a prediction region, reports its coverage and its size, and stops. Two things are invisible from that vantage point. First, a region’s effect on a downstream control decision — specifically, whether it is used to trigger a fallback — is not evaluated at all; open-loop coverage and closed-loop safety outcome are simply assumed to move together. Second, and more consequentially, *triggering the fallback is not free*. A vehicle that stops in a live lane because its uncertainty region grew large is not obviously safer than one that continues; an evaluation that reports only coverage, or only collision rate under the fallback, cannot see the trade-off between the two, and cannot see that a policy which triggers whenever it is asked to would trivially achieve a good miss rate while being useless.

This paper studies fallback triggering as a decision problem with a formal guarantee attached to the decision itself, evaluated where that guarantee actually has to hold: in closed loop, on real driving scenarios, with a trigger threshold that has consequences for what the vehicle does next. We make three contributions.

- 1) We formalise minimum-risk-manoevrue triggering as *selective prediction* with a guarantee on the closed-loop miss rate — the probability that a harmful event occurs without the fallback having fired first — rather than on marginal coverage of a static prediction region. We calibrate the trigger threshold with Learn-then-Test [2], which requires no assumption that the closed-loop loss be monotone in the threshold, and report Conformal Risk Control [3] as a tighter alternative on the scenarios where monotonicity is observed to hold.
- 2) We evaluate entirely in closed loop, on scenarios replayed from real driving logs (Argoverse 2 and nuScenes, via ScenarioNet [8] and MetaDrive) under reactive traffic, and we report the unnecessary-stop rate

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alongside the missed-intervention rate at every operating point — the over-conservatism trade-off that an open-loop study structurally cannot see.

- 3) We connect the closed-loop outcome to the underlying predictor’s calibration under dataset shift, asking directly whether a degradation in open-loop conformal coverage translates into a measurable degradation in closed-loop safety, and report the answer honestly whether or not it supports the premise.

The remainder of the paper is organised as follows. Section II situates this work against open-loop conformal prediction for driving, task-relevant failure detection, and closed-loop safety evaluation more broadly. Section III formalises the triggering problem and the calibration procedure. Section IV describes the closed-loop evaluation environment, the fallback manoeuvre, and the harm labelling used to score every rollout. Section V reports results. Section VI discusses limitations, and Section VII concludes.

II. RELATED WORK

A. Conformal prediction for autonomous driving

Conformal prediction [1] constructs prediction sets with a distribution-free, finite-sample marginal coverage guarantee under exchangeability between calibration and test data, and has been applied extensively to trajectory forecasting and to safe planning under uncertainty. Lindemann *et al.* embed conformal prediction regions directly into a model predictive controller to obtain probabilistic safety guarantees, but their guarantee collapses once the deployment distribution shifts away from the calibration distribution — a limitation Rahaman *et al.* [4] address with robust conformal prediction driven by a conditional generative model of the shift, evaluated in a synthetic multi-agent corridor. SafePath [5] integrates conformal prediction into an LLM-based path-planning pipeline, providing a guarantee that a set of candidate paths contains a safe one with a user-specified probability, and reports both open-loop (nuScenes) and closed-loop (highway-env, synthetic traffic) results. In all three cases the guarantee is on the *prediction* — a region contains the truth, or a path set contains a safe path — rather than on the eventual control decision the region is used to make, and none evaluates a fallback-triggering decision on real, replayed driving logs under reactive traffic. This paper’s guarantee is instead on the decision: the probability that a minimum-risk manoeuvre fires before a harmful event, evaluated end-to-end in closed loop.

B. Task-relevant and closed-loop failure detection

Farid *et al.* [6] propose a run-time monitor for trajectory-prediction failures that is *task-relevant* by construction: it flags a misprediction only when the resulting planning cost is anomalous relative to the predicted cost distribution, with data-free false-positive and false-negative rate bounds. Their detector is evaluated against hand-labelled scenarios in an open-loop and semi-reactive setting, and — like the conformal-prediction literature above — does not close the loop through an actuated fallback and re-measure the outcome. Their central insight, that a task-irrelevant misprediction should not count against a

detector, is reflected in this paper’s harm labelling: a collision the ego vehicle suffers while already stopped or while being struck from behind is scored as a distinct *induced-collision* cost of the fallback itself, never as a missed intervention.

C. Risk-sensitive safety analysis

Chapman *et al.* [7] develop risk-sensitive safe sets using Conditional Value-at-Risk to assess the severity, not merely the probability, of constraint violation along a stochastic trajectory, providing the risk-sensitive vocabulary this paper borrows in motivating why a mean missed-intervention rate is the appropriate quantity to control at a chosen operating point, even though a tail-severity extension (Section VI) is left to future work.

D. Closed-loop simulation infrastructure

ScenarioNet [8] unifies heterogeneous real-world driving logs — including Argoverse 2, nuScenes, nuPlan, and Lyft Level 5 — into a common scenario description that can be replayed interactively in the MetaDrive simulator, with either logged-trajectory replay or reactive rule-based (IDM) traffic. This work is built directly on that infrastructure: every scenario reported in Section V is a real logged driving scenario, replayed headless with reactive traffic, not a synthetic or hand-crafted one.

III. PROBLEM FORMULATION AND METHOD

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ : at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor’s output for nearby agents, and engages the minimum-risk manoeuvre the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{I}[\text{harm in } X \text{ under } \lambda, \text{ MRM not first}], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{I}[\text{MRM fires under } \lambda, X \text{ harm-free at } \lambda = \infty], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1, \dots, X_n} \left(\mathbb{E}_X [L(X, \hat{\lambda})] \leq \alpha \right) \geq 1 - \delta, \quad (3)$$

while minimising $\mathbb{E}_X [S(X, \hat{\lambda})]$ — the smallest autonomy loss compatible with the risk target α .

Equation (3) is not a conformal coverage statement: $L(X, \lambda)$ depends on the entire closed-loop rollout under λ , since engaging the fallback changes every subsequent state, so it is neither a marginal coverage event on a static prediction nor guaranteed to be monotone in λ a priori. We therefore calibrate $\hat{\lambda}$ with Learn-then-Test [2]: for a finite grid $\lambda_1 > \lambda_2 > \dots > \lambda_J$ ordered from least to most conservative, compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n} \sum_i L(X_i, \lambda_j)$ on n

calibration scenarios, a Hoeffding–Bentkus p -value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, and apply fixed-sequence testing starting from the most conservative λ ; every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ , and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). Where the empirical loss is observed to be monotone in λ across the calibration set, we additionally report Conformal Risk Control [3], whose expectation guarantee is tighter but requires that monotonicity assumption; where it is violated we report the violation rate directly rather than silently falling back to the weaker method.

B. Trigger score

At each decision tick the trigger score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle’s own near-term extent. Writing the ego’s constant-heading, constant-speed extrapolation over a horizon of T steps as $\{p_\tau^{\text{ego}}\}_{\tau=1}^T$ — the same route-agnostic motion model used by the harm metric in Section IV-B, deliberately *not* the vehicle’s planned, route-following path, so that the trigger cannot credit itself for steering behaviour the safety label does not assume — and a predictor’s K -mode forecast for agent i as $\{\hat{p}_\tau^{(i,k)}\}$, each predicted offset is decomposed into a longitudinal component $a_\tau^{(i,k)}$ (along the ego heading) and a lateral component $\ell_\tau^{(i,k)}$ (perpendicular to it), each compared against a separate margin: a length-based longitudinal margin $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and a width-based lateral margin $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion at (i, k, τ) is

$$\rho_\tau^{(i,k)} = \max \left(0, 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}} \right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}} \right)^2} \right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. A single isotropic clearance radius cannot represent both margins without either missing a straight-ahead lead vehicle (a radius set by width alone) or over-triggering on adjacent-lane traffic (a radius set by the vehicle’s half-diagonal); the decomposition in (4) is required for correctness, not merely convenient, a fact confirmed empirically during development (Section IV).

C. Fallback manoeuvre

On triggering, the vehicle executes the minimum-risk manoeuvre defined by UN Regulation No. 157 (Automated Lane Keeping Systems) §5.5.1 [10]: decelerate inside the current lane with a deceleration demand not exceeding 4.0 m/s^2 , lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0t, 0)$ from the moment of triggering.

IV. EXPERIMENTAL SETUP

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [8] from Argoverse 2 Motion Forecasting and nuScenes. Non-ego traffic is controlled

reactively (rule-based, IDM), not replayed open-loop from the log: pure log replay was found to roughly triple the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its logged trajectory — precisely the deviation a fallback manoeuvre produces. Reporting results under non-reactive replay would therefore misrepresent the cost of stopping, which is a central quantity this paper reports.

B. Harm labelling

A rollout is labelled harmful using the nuPlan closed-loop safety definition [9]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary and not evaluated against tracks behind the ego. Fault attribution follows the nuPlan convention: a collision the ego suffers while stationary, or while being struck from behind, is *not* at-fault and is never counted as a missed intervention; it is instead recorded as an *induced collision* — the concrete, measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot, run online inside the closed loop so that its input reflects the actual — possibly ego-deviated — simulator history rather than the original log, which is necessary once traffic is reactive. [Predictor training data, checkpoint selection, and validation minADE to be finalised in the camera-ready version alongside Section V.]

D. Splits and pre-registration

Design decisions were made on a held-out development split, disjoint from both the calibration split used to select $\hat{\lambda}$ and the test split used only for the reported evaluation. The harm definition, risk level α , confidence level δ , and the refutation criteria for every hypothesis below were fixed and recorded before any calibration- or test-split scenario was evaluated.

V. RESULTS

Results in this section are pending the full-scale calibration and test-split evaluation ($n \approx 2,000$ scenarios per split) and are not yet reported. Diagnostic runs at small scale ($n \leq 300$) were used exclusively to validate the pipeline described in Sections III–IV — in particular, to identify and correct two errors in the trigger-score geometry of (4) before any pre-registered evaluation — and are not reported here as results.

A. H1: does open-loop ranking predict closed-loop ranking?

[Pending.]

B. H2: does calibrated triggering dominate hand-tuned thresholds?

[Pending.]

C. H3: does open-loop coverage collapse translate to closed-loop harm?

[Pending.]

VI. DISCUSSION

[To be completed alongside Section V. Planned content includes: sensitivity of $\hat{\lambda}$ to decision latency and decision rate; the CVaR-style tail-severity extension motivated by [7], replacing the mean missed-intervention rate in (3) with a tail bound on a continuous time-to-collision shortfall; and the limits of the reactive-IDM traffic model as a proxy for real driver response to a decelerating vehicle.]

VII. CONCLUSION

[To be completed once Section V is populated.]

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REFERENCES

- [1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Berlin, Germany: Springer-Verlag, 2005.
- [2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, “Learn then test: Calibrating predictive algorithms to achieve risk control,” *arXiv preprint arXiv:2110.01052*, 2021.
- [3] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” *arXiv preprint arXiv:2208.02814*, 2022.
- [4] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, “When environments shift: Safe planning with generative priors and robust conformal prediction,” in *Proc. 8th Annu. Conf. Learn. Dyn. Control (LADC)*, 2026.
- [5] A. Doula, M. Mühlhäuser, and A. S. Guinea, “SafePath: Conformal prediction for safe LLM-based autonomous navigation,” *arXiv preprint arXiv:2505.09427*, 2025.
- [6] A. Farid, S. Veer, B. Ivanovic, K. Leung, and M. Pavone, “Task-relevant failure detection for trajectory predictors in autonomous vehicles,” in *Proc. 6th Conf. Robot Learn. (CoRL)*, Auckland, New Zealand, 2022.
- [7] M. P. Chapman, R. Bonalli, K. M. Smith, I. Yang, M. Pavone, and C. J. Tomlin, “Risk-sensitive safety analysis using conditional value-at-risk,” *IEEE Trans. Autom. Control*, vol. 67, no. 12, pp. 6521–6536, Dec. 2022.
- [8] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, “ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling,” in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS), Track Datasets and Benchmarks*, 2023.
- [9] N. Karnchanachari *et al.*, “Towards learning-based planning: The nuPlan benchmark for real-world autonomous driving,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [10] United Nations Economic Commission for Europe, “UN Regulation No. 157 — Uniform provisions concerning the approval of vehicles with regard to Automated Lane Keeping Systems,” Rev. 1, 2025.